

Aurora v1 — AI SoC

A complete System-on-Chip taken from RTL to a signed-off GDSII layout, using only open-source tools, on a single 23 GB laptop.

Engineering report · Yashwanth Kudum · 2026

1. Executive summary

Aurora v1 is a multi-clock AI System-on-Chip integrating a 64-bit RISC-V control core with a hardware matrix-multiply accelerator over a coherent-terminated AXI4 fabric. It was carried through the **entire physical-design flow** — RTL, lint, simulation, formal, synthesis, floorplan, placement, clock-tree, routing, timing, DRC, LVS — to a manufacturable GDSII layout on the SkyWater Sky130B process. Final sign-off: **timing MET** across all process corners, **DRC = 1** (fill waiver), and a **device-perfect LVS match**.

RTL → GDSII complete · 137,562 devices · ~1.97 M instances · 41.25 mm² die · 33/50 MHz multi-clock

2. Architecture

A Rocket RV64IMAC tile drives the fabric through dual AXI4 masters (cacheable memory + uncached MMIO), clock-crossed from the 33 MHz CPU domain into the 50 MHz fabric. An 8×8 round-robin AXI4 crossbar routes to a boot ROM, on-chip SRAM, UART, GPIO, timer, and the tensor accelerator.

**Rocket RV64IMAC
CPU tile (macro)
33 MHz**

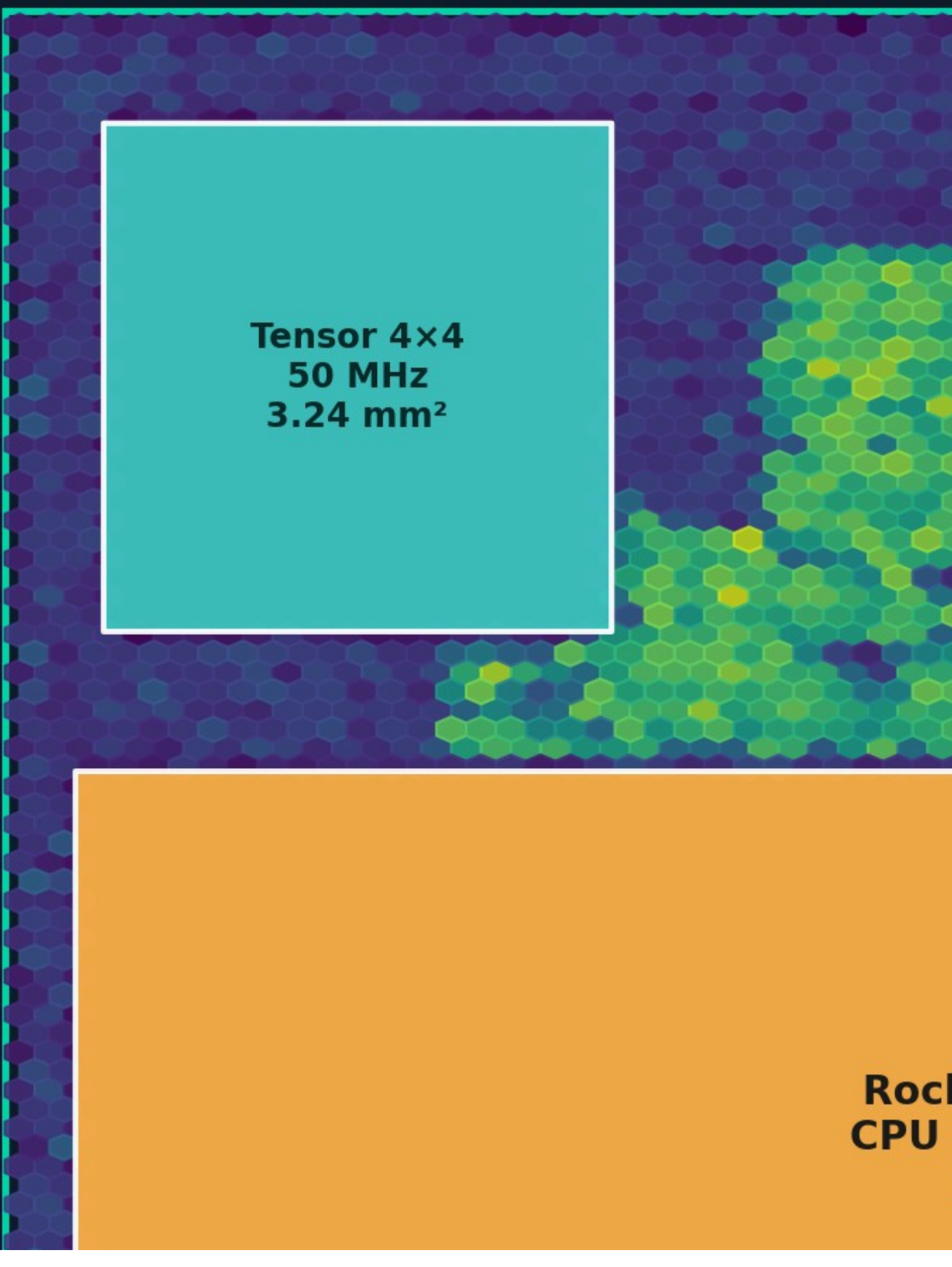
**AXI
CDC**

mem_axi4 + mmio_axi4

3. Physical layout

The Rocket CPU (18.2 mm²) anchors the bottom of the die; the tensor accelerator (3.24 mm²) and the std-cell fabric occupy the upper region. The figure below is rendered from the actual placement database — the colored sea is the ~130,500-cell glue logic.

Aurora v1 — Full-Chip Lay



The diagram shows a full-chip layout for Aurora v1. It features a dark blue background with a hexagonal grid pattern. A large cyan rectangle is positioned in the upper left, and a large orange rectangle is in the lower right. The cyan rectangle contains text about a tensor core, and the orange rectangle contains text about a Rockchip CPU. A vertical cyan line is on the left side of the cyan rectangle.

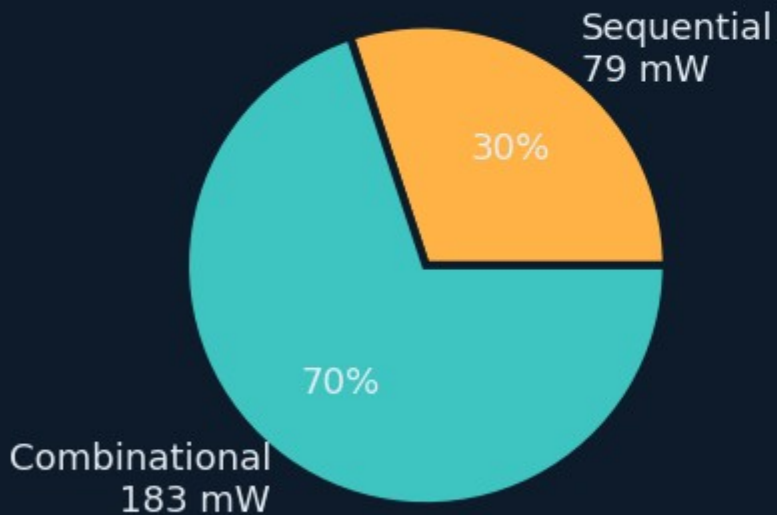
Tensor 4×4
50 MHz
3.24 mm²

Rockchip
CPU

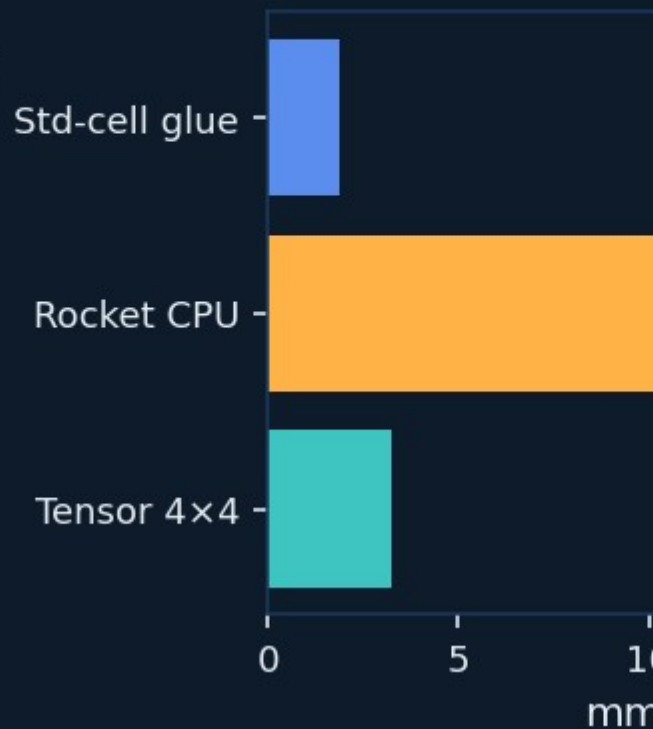
4. Sign-off results

Aurora v1 — Sig

Power — 263 mW (fabric)*



Area on 41.2



complete

137,562

~1.97 M

RTL→GDSII

Devices

Instances

* fabric/glue + clock-tree power; macro i

Check	Result
Detailed routing (TritonRoute)	converged — 101 residual
Multi-corner post-route STA (SPEF)	WNS / TNS = 0.00 — MET at ss / tt / ff
DRC (Magic)	1 — isolated nwell fill tap (waiver)
LVS (netgen)	device-perfect match — 137,562 = 137,562
GDSII	streamed — full-chip layout (1.32 GB)

Honest scope: 263 mW power is fabric/glue + clock-tree only (macro internals not in the abstract power models). LVS shows a 2-net delta from 4 deliberately-unused AXI lock outputs and the VPB/VNB body-pin extraction convention — benign waiver-class items, not logic defects.

5. Functional verification

The full SoC boots in simulation: the Rocket core comes up through the real fabric, initializes operand matrices, launches the tensor core, polls for completion, reads the result back over AXI, and reaches its run loop — zero traps.



full-SoC boot simulation

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AURORA v1 AI SoC — BOOT

1x Rocket RV64IMAC + 4x4 Tensor

fabric 50 MHz · CPU 33 MHz · Sky130B

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[1] Initializing matrices in SRAM...

[2] Running matrix-multiply on Tensor Core

[3] DONE! result read back over AXI

[4] Aurora Tensor Core: OPERATIONAL

Rocket booted through the real AXI fab

Aurora v1 AI SoC is ALIVE!

6. Engineering highlights

- **Cracked a routing-congestion wall** on the dense systolic datapath — diagnosed a metal-1 short-storm and beat it with a soft per-layer routing-resource derate.
- **Hardened a RISC-V tile as an AXI macro** — re-wrapped Rocket's TileLink-C port through a cache cork + TL-to-AXI4 bridge so it presents a clean AXI4 master.
- **Full-chip PDN/LVS integration** — root-caused a power-grid fragmentation (a macro

exposing power on two metal layers) and fixed it with a PDN keep-out so the power net extracts as one node.

- **Memory-bounded P&R** — the entire flow runs capped at 18 GB to survive on a 23 GB host, with resume-from-checkpoint to iterate the heavy steps.

7. Specifications

Parameter	Value
Process / library	SkyWater Sky130B · sky130_fd_sc_hd · 5 metal layers
Die area	7500 × 5500 μm (41.25 mm²)
CPU	Rocket RV64IMAC, 33 MHz, dual AXI4 masters
Accelerator	4×4 systolic int16 MAC array, 50 MHz, AXI4-Lite
Fabric	128-bit 8×8 AXI4 crossbar, 50 MHz
Toolchain	Verilator · Yosys · sv2v · SymbiYosys · OpenLane (OpenROAD/Magic/netgen/KLayout)

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github.com/kudumyashwanth/aurora-v1 · Figures from the chip_route2 sign-off run.